Urban Heat Islands and Heatwaves in Megacities

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1-Project Overview

Rapid urbanization creates urban heat islands, causing megacities to experience significantly higher temperatures than their rural surroundings. This project focuses on analyzing the relationship between urban land use and rising surface temperatures. Various data visualization techniques will be used to analyze temporal temperature shifts and spatial disparities across metropolitan regions. The main goal is to raise awareness about the direct impact of urban planning on climate vulnerabilities in line with SDG 11 and SDG 13.

2-Dataset Information

- **Dataset Names:**
  1. ERA5-Land Monthly Averaged Data (Surface Temperature)
  2. Urban Population Indicators
  3. Global Human Settlement Layer (GHSL) - Built-up Surfaces
  4. MODIS Vegetation Indices (NDVI - Green Space Tracking)
- **Data Providers:** ECMWF, The World Bank, European Commission (JRC), NASA
- **Data Sources:** Copernicus Climate Data Store, World Bank Open Data, JRC Data Catalogue, NASA Earthdata
- **Dataset Links:**
  - https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land-monthly-means
  - https://data.worldbank.org/indicator/SP.URB.TOTL
  - https://ghsl.jrc.ec.europa.eu/datasets.php
  - https://search.earthdata.nasa.gov/search?q=MOD13Q1